BIO204 FINAL TERM NOTES

BY MEHWISH MUGHAL

GROUP:
BIOTECH
BRAINY BUNCH

Email:
mehwishshabirhussain07@gmail.com

Scanned with CamScanner

Bio204 Final Term Notes

Oxygen Delivery System:

04 parts:
- commonly used pressure (250kPa) – oil free air
1. Compressor → generate pressure to force the air into sparger holes
2. inlet air sterilization system → prevent contamination entry
3. Air sparger.
4. Exit air sterilization system

prevent entry of contamination and also prevent organisms in the reactor to contaminate the air

Inlet and exit sterilization system comprise of filtration

(≤ 5 litres volume)

disk shaped hydrophobic Teflon in a polypylene housing are used.

(tough and reusable)

upto >10,000 Litres

(upto 1000 litres)

Pleated membrane filters housed in polypropylene cartridges are used

(create compact factor by increasing filteration area and decreasing pressure)

Heat sterilization (steam used)

used cz' filteration is too expensive and its difficult to achieve energy requirements

otherwise HEPA filters used

Scanned with CamScanner

Condenser → heat exchanger which passes cold water around
- condenses water vapours and volatile materials on the inner condenser membrane.
- reduces evaporation and loss of volalites

Q: How to maintain proper aseptic condition in fermenter?

① Maintain positive pressure

② If pressure is not maintained, air is not continously pumped into the reactor, a vacuum will form and causes contamination.

③ Pressure inside reactor must be greater than pressure outside

Sparger → breaks incoming air into small bubbles.
- commonly used design is sparger ring.

Sparger Ring
- hollow tube with small holes
- easy to clean and less likely to blockage
- located below agitator and same diameter as impeller

Close air feed valve when emptying fermenter.
- reduces risks of contamination.

Bubbles → directly beneath impeller

Shear forces high around impeller
- maximizes bubbles efficiency to break
- determine bubble size

impeller speed slow
↓
Bubbles not broken

impeller speed high
↓
equal bubbles distribution

Scanned with CamScanner

Formula for airflow rate:-

$$\text{Air flow rate} = \frac{\text{volumetric airflow rate}}{\text{liquid volume}}$$

units = volume per volume per minute (vvm)

Air and liquid must be in litre

Amount of dissolved oxygen in fermenter is controlled by two methods:

1. Increased flow rate
2. Increased agitation

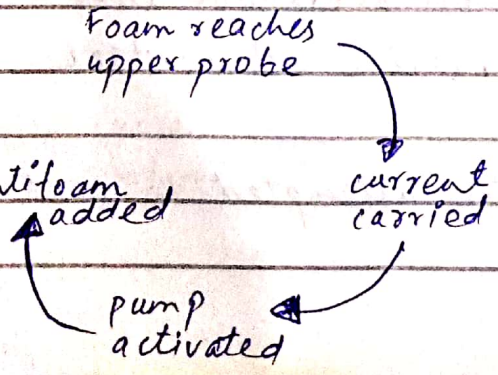
FOAM CONTROL SYSTEM:-

Disadvantages of Excessive foam production:

1. Blocked air exit filters
2. Pressure build in reactor
3. Loss of medium
4. Damage to reactor
5. Injury to operating personnel
6. Poor oxygen transfer rates

Detection of Foam

- two conductivity/level probes are used
- one probe immersed in liquid
- 2nd probe above the liquid level



Antifoams

Foam can be controlled by adding:

1. Antifoaming agents based on silicon or vegetable oils.

there are six requirements for antifoams

1. nature of medium
2. Products produced by fermentation
3. Aeration rate and stirrer speed
4. Use of mechanical foam controllers
5. Head space volume
6. condenser temp

Scanned with CamScanner

pH control System:-

- controlled by neutralizing agents
  - these agents should be:
    1. non-toxic
    2. non-corrosive

KOH — less toxic → NaOH

NaOH — less expensive → KOH

HCl — corrosive → not used

H_2SO_4 — used → other than the range of 10-80%

1. added to fermenter using peristaltic pump
2. Silicon tubing also used
3. Tygon or Neoprene tubing also used

Tygon → non-autoclavable

Neoprene → autoclavable

- Indeed not all fermentations need pH control system

Automatic System of pH control

- 3 parts:
  1. pH probe → steam sterilizable (highly sensitive to temp)
  2. alkali delivery system → turns on when pH is low
  3. acid delivery system → turns on when pH is high

designed to have dead-band

dead-band is used to prevent excessive acid-base addition

If dead band too small → pH overshoot and undershoot

Dead band lead to less precise pH control

Set-point:-

- pH at which fermenter is attempted to controlled

Scanned with CamScanner

Cleaning & Sterilization Facilities:-

Small scale reactors → cleaned and auto-claved
large scale reactors → clean in place (CIP).

CIP → cleaning of fermenter + all linked pipelines
- → steam, cleaning & sterilizing chemicals, spray balls used
- → automatic process (no manual error)

Sterilization → process to prevent contamination
If contamination takes place; following situations occur:
① Medium support product + contamination = loss in productivity
② If continous fermentation → contamination displace product
③ foreign organism contaminate final product
④ Contamination makes the extraction of product difficult.
⑤ Contamination degrade desired products
⑥ Contamination result in lysis of culture

Definition of Sterilization
- → removal or destruction of all micro-organism
- → sterilizing agent is called sterilant.

liquids / Gases → sterilized by filteration

Source of Contamination
① leakage of vessel
② Actual medium
③ Air sparger system can contain contaminant

How to avoid contamination
① Using pure inoculum
② Sterilize medium
③ Sterilize vessels
④ Sterilize all materials
⑤ Maintain asaptic conditions

Scanned with CamScanner

Sterilization Kinetics:-

Media sterilization → filteration / radiation / Heat / chemical / Ultrasoni

$-\frac{dN}{dt} = kN$ → first order reaction

Taking integaration

$\ln(N_t/N_0) = -kt$

Graph b/w N_t/N_0 against time → straight line

Straight line gives $-K$

short Q

Kinetic description makes two ~~prediction~~ predictions:

① Infinite time required $N_t = 0$

② After certain time, viable cell < 1

* K is not only specie dependent, but also physiological form of cell
* reaction increases with increase in temperature due to increase in reaction rate constant.
* K is constant; only if temp is constant.

Relationship b/w K and temp

Arrhenius equation

$$\frac{d \ln K}{dT} = \frac{E}{RT^2}$$

Taking integaration

$$K = Ae^{-E/RT}$$

Taking natural log

$$\ln K = \ln A - \frac{E}{RT}$$

combining eq.

$$\ln \frac{N_0}{N_t} = A \cdot t \cdot e^{-E/RT}$$

Graph b/w $\ln K$ and $\frac{1}{T}$

↓

Straight line

↓

Arrhenius plot

↓

used to calculate activation energy and prediction of reaction rate

Scanned with CamScanner

Deindoefer & Humphrey (1959)

→ $\ln \frac{N_0}{N_t}$ → del factor / Nabla factor (∇)

Del factor

↳ measure of fractional reduction in viable organism count produced by a certain heat and time.

formula $\nabla = \ln\left(\frac{N_0}{N_t}\right)$

$\ln\left(\frac{N_0}{N_t}\right) = kt$

$kt = A \cdot t \cdot e^{-(E/RT)}$

$\nabla = A \cdot t \cdot e^{-(E/RT)}$

Rearranging equations

$$\ln t = \frac{E}{RT} + \ln\left(\frac{\nabla}{A}\right)$$

Fermentation medium is not inert

↓

temperature of sterilization causes nutrient loss

2 types of reaction contribute to heat loss:

↓

① Interaction b/w nutrients and medium

② Degaradation of heat labile components

Thermal destruction of essential media components:-

→ $\frac{x_t}{x_0} = e^{-kt}$

Effect of temperature on reaction rate constant:

↳ Arrhenius Equation

$$\ln K = \ln A - \frac{E}{RT}$$

∴ E = activation energy

R = Gas constant

Graph b/w ln K and 1/T gives $-\left(\frac{E}{R}\right)$ → straight line graph

Bacillus stearothermophilus

↳ Indicator for fermentation

Scanned with CamScanner

Continuous stream

High temp → high sterilization → short time → low nutrient loss

Sterilization

- In-situ (in place) — Large scale
- Ex-situ (out of place) — small scale

Modern Continuous Sterilizers

↓

Double spiral heat exchangers which separate two streams

↓

low chances of blockage

↓

precision through computers

Advantages of Continuous sterilization over Batch sterilization:-

1. Superior maintaince Quality
2. Ease of scale-up
3. Automatic control
4. reduction of sugar capacity for steam
5. reduction of cycle time
6. reduction of corrosion

Advantages of Batch Sterilization over continuous sterilization

1. low costs
2. low contamination
3. Easy manual control
4. Easy to use with solid media

Unsuitable conditions of Early continuous Sterilizers:-

1. Failure of gaskets b/w plates results in mixing of sterile and un-sterile medium

2. Particulate components in media block heat exchangers

Scanned with CamScanner

Media Sterilization in Batch Mode:-

→ more nutrient loss than continous sterilization → maximum feasible temperature 121°C → time should be as minimum as possible

Deindoerfer & Humphrey (1959)

→ requirements for batch sterilization

↓

① profile of increase and decrease in temperature

② No. of organisms in medium should be known

③ Thermal death characteristics of design organisms.

Richards (1968) Graphical Representation

↓

time divided into increments

t_1, t_2, t_3, etc (x-axis)

temperature (y-axis)

For each point;

$\nabla_1 = k_1 t$, $\nabla_2 = k_2 t$, $\nabla_3 = k_3 t$ etc

For heating pattern;

$\nabla_{heating} = \nabla_1 + \nabla_2 + \nabla_3$

same for cooling pattern

If total no. of organisms is known

↓

we can calculate the Del factor

Specific case

$\nabla = \ln(10^{n}/10^{-3}) = 32.2$

Overall Del factor formula

↓

$\nabla_{heating} + \nabla_{cooling} + \nabla_{holding}$

(temperature 121°C)

Relation b/w Del factor, time and temperature

↓

$\nabla = A \cdot t \cdot e^{-(E/RT)}$

as temp and time varry in batch mode; there are different linear, exponenti and hyperbolic curves instead of straight line

Scanned with CamScanner

Holding Del factor at constant temp

$\nabla_{holding} = \nabla_{overall} - \nabla_{heating} - \nabla_{cooling}$

As $\nabla = kt$

when we know holding temp and del factor we can calculate time by; $t = \nabla / k$

Holding time is calculated by;

$t = \nabla_{overall} / k$

Banks (1979)

→ discussed use of del factor in scale up of the batch sterilization

Del factor does not dependent on volume term.

increase in size of fermenter → increase the del factor

large fermenter size → require more holding time at the same temp which results in nutrients loss.

Rapid Method for sterilization cycle by Richards (1968)

→ assume all sterilization occur above 100°C and heating cooling parts are linear.

(Both assumptions Valid)

→ this technique is simple and loose less accuracy

(temp increase by 1°C and del factor gradually increase)

In autoclave;

temperature = 121°C

specific death rate = 2.538

del factor = 42.549

Imp condition

Methods of Medium Sterilization in Batch-Culture :-

- fermentation vessel
- separate mash cooker

Advantages

One cooker → serve several fermenters and medium is sterilized

- concentrated medium is sterilized in cooker and diluted form is used in fermenter
- in case of more viscous medium, less power motor can be used.
- prevents corroison in fermenter

Disadvantages of Separate Cooker :-

1. same cost of cooker as that of sterilizer
2. Complex pipeline with inherent dangers of contamination.
3. Mechanical failure causes huge medium loss of several fermenters.

Overview

Separate mash cooker is considered costly and contain high risks of contaminatio

↓

It is a method of choice

Scanned with CamScanner

Design of Continuous Sterilization Process:-

Temperature raised by continuous heat exchanger

Temperature maintained by insulated serpentine holding coil

length of holding period dictated by length of coil + flow rate of medium

cooling period acheived by

{ two sequential heat exchangers }

- uses incoming media as a cooling source
- uses water as a cooling source

Types of Continous Sterilizers

- Indirect heat exchanger
- Direct heat exchangers (steam injector)

Major advantage

Higher temperature is utilized by reducing the holding time and reducing the nutrient loss.

Del factor is acheived by addition of temp and holding time

Heating and cooling periods are very small as compared to batch sterilization

Scanned with CamScanner

→ both direct and indirect heat exchangers used for starch-containing broths.

Indirect Heat Exchangers for continous Sterilization

Double-Spiral Type

consisting two sheets of high grade stainless steel curved around centre. Ends covered by cover.

Method

Steam and medium move in opposite coils in opposite directions. Unsterile medium is used as cooling agent.

Advantages

① two steams separated by stainless steel barrier with gasket seals at end of plates which makes cross contamination

② Spiral route transvered by medium allows more clearance and reduces the risks of sedimentations, burning, and fouling on.

Direct Heat Exchangers for continous sterilization

Counter Plates Type;

plates separated by gasket and injector injects steam directly into unsterile broth. Extra plates addition increase its capacity

Advantages

① short method
② use media with suspended sol
③ low cost
④ easy cleaning & maintence
⑤ High steam utilization
⑥ more adaptable than spiral type

Disadvantages

① foaming may occur
② Requires allowance for condense dilutions
③ Require clean steam
④ should be free from anticorrosion agents.

Scanned with CamScanner

Sterilization of Fermenter:-

heating of jacket or coil of fermenter with steam and sparging steam.

Steam pressure = 15psi
time = 20min
temperature = 121°C

Sterilization of Feed:-

additives should be sterilized on the basis of their nature, volume, and feed rate.

large volume of feed → continous sterilization

Batch sterilization of feed → steam injection

Filtration Mechanisms:-

1. inertial impaction
2. Diffusion
3. Electrostatic attraction
4. Interception

Sterilization of liquid Wastes:-

Strict containment regulations

* May be batch / continous sterilization under contained conditions

• Batch Sterilization → Steam
Continous " → heat exchanger

Temperature should be atleast 60°C before wastes discharge.

Filter Sterilization

2 types of filters

- filter pores smaller than particles
- filter pores larger than particles

Ideal filteration System for media sterilization:-

1. media should be free of fungi, bacteria & mycoplasma
2. minimal adsorption of protein
3. Free of viruses
4. Free of bacterial endotoxins

Scanned with CamScanner

Inertial Impaction

- significant in gases

Method:-

- suspended particles in fluid have momentum and move through filter in straight lines via the route of least resistance.

Diffusion

- significant in gases

Method:-

Extremely small particles undergo Brownian motion and start collision. Thus; impact on filter

Electrostatic Attraction

- attraction b/w opposite charged ions.

Interception

- comprise of fibres with define openings of various size

Method

- Particles larger than pore size removed through interception

Filter Sterilization of Media:-

designed for the mediums which contain heat laible components

Manufactured filters consist of:

① membrane cartridges fitted into stainless steel

② membranes made of steam sterilizable hydrophillic material

Pre-filters required to prevent the blockage of filters

Filter Sterilization of Air:-

- fixed pore filters made of pleated membrane cartridges

Polytetrafluoroethylene (PTFE)

- used in pleated membranes
- hydrophobic in nature; thus; resistance to wetting

PTFE is resistance to ammonia which is added for pH control

Coalesced filters → pre-filtering

- Automatic Drain

Scanned with CamScanner

Sterilization of fermenter Exhaust air:-

↳ designed for the advent use of recombinant organisms

Fixed pore membranes used at high temp with large containment level

↓

foam may overflow the fermenter into exhaust air

Prefilter → hydrophobic or mechanical sparger

Recovery and Purification of Fermentation Products:-

↓

Difficult and costly process

cost range → 15% - 70% of the total cost

At time of harvesting media contains:

1. Intact-microorganisms
2. Cell fragments
3. Metabolic Products
4. Soluble and insoluble media components.

Product → Intracellular & heat laible

corrected Equipment type selected for harvesting product at a limited time

Recovery Process Criteria

1. Product may be intracellular or extracellular
2. Concentration of product in media
3. Physical & chemical properties
4. Intended use of product
5. Minimal acceptable standard of purity
6. Magnitude of bio-hazard
7. impurities in fermenter broth
8. Marketable price of product.

Main objective at 1st stage

↓

removal of large solid particles and microbial cells by centrifugation / filteration

↳ broth is fractioned using various methods

↳ chromatography and crystallization to obtain concentrated and impurities free product

Scanned with CamScanner

Techniques for handling Broth:-

1. selection of micro-organisms which produce metabolites
2. Modification to reduce production of metabolites
3. Precise timing of harvesting
4. pH control after harvesting
5. temperature control after harvesting
6. Use of enzymes to attack cell walls
7. Addition of flocculating agents

Requirements for Recovery and Purification

1. capital cost
2. Processing costs
3. Throughput requirements
4. Yield Potential
5. Product Quality
6. Technical Expertise available

Solvent-recovery Plant
→ Usually distillation

Advantages of centrifugation over filteration:-

1. Filteration → slow/difficult
2. → necessary to obtain free filter-aids
3. Hygeine required in continous

Centrifuge is made of perforated bowl lined with filter bag of nylon/cotton

Basket centrifuge
→ Separation of Mould mycella or crystalline compounds

Interacellular Product
→ cell disruption

chemical
1. Detergents
2. Osmotic Stock
3. Alkali treatment
4. Enzyme //

Physio-mechanic
1. liquid shear
2. solid shear
3. Agitation
4. Freeze-thawing
5. Ultrasonication

Scanned with CamScanner

Drying of Product:-

① reduced cost of transport
② Easier to handle & package
③ can be stored easily in dry form.

Spray Drier → feed below 6kg/min

Freeze Drying

- no harm to heat laibles
- important in operation and production of many biologicals and pharmaceuticals.

Crystallization

- used in initial recovery of organic acids and amino acids
- ① considered due to simplicity

Effluent

- whole waste material of fermentation process

In-situ recovery by Rotfiltrate

① Vaccum and flash fermentation of ethanol
② Extractive fermentation for organic acids & toxins
③ Adsorption for ethanol and cycloheximide
④ Ion-exchange in antibiotics and salicyclic acid
⑤ Dialysis for lactic acid, salicyclic acid & cycloheximide

Typical wastes

- unconsumed organic & inorganic components, microbial cells, suspended solids, filter aids, waste water, traces of solvents, acids, alkalis, human sewage

Sewage Treatment Work (STW)

- for the disposal of liquid wastes

Fermentation effluent do not affect aquatic flora & fauna.

Scanned with CamScanner

Biochemical Oxygen Demand (BOD)

- measure of Quantity of oxygen required for oxidation of organic matter in water, by the micro-organisms present in given time interval, at T temp.

level of BOD in effluent
- give standard for the effluent treatment

Factors to investigate site survey

1. BOD/COD
2. pH range
3. Turbidity
4. Temp range
5. Color
6. Hardness
7. Detergents
8. Radioactivity
9. Odour & Taste
10. Daily flow rate

Disposal of Effluents to sewers:-

- Requirements:

1. capacity to cope with
2. no interferance with process of treatment.
3. No compounds present in effluents which cause problems

Effluent Treatment Program for a site:-

1. Water reused
2. Concentrated waste streams recovered as food, feed, fuel.
3. Toxic effulents treatment.
4. effulents loading under maximum productions
5. Effulents discharge into municipal sewers.
6. Effulents discharge directly: if no risks available.

Effulents Disposal Methods

1. effulents discharged into land, river, sea
2. disposal in a lantill site, or increarated.
3. Partially pre-treatment
4. Remainder treated at sewage work before discharge.
5. All effulent sent at sewage works
6. All effulent treated at factor

Scanned with CamScanner

Types of Fermentations

waste treatment process:-

1. Physical treatment } Primary Treatment (sedimentation and coagulation)
2. Chemical treatment }
3. Biological treatment → secondary treatment

Physical / Chemical / Biological → Tertiary Treatment
(used to improve Quality)

Sludge conditioning then disposed by landfilling / incineration

Aerobic Process

- Trickling Filters → fixed film bioreactor
- Rotating Disc Contractor
- Activated sludge processes → Aeration & Agitation
- Biological Aerated Filters (BAFs) → improved version of trickling filters

Anaerobic Process

- Digestion
- Filteration
- Sludge Blankets

Advantages of Anaerobic Process

1. higher loading rates
2. low requirements of BOD treated
3. Useful end-products
4. Organic matter metabolize to stable form.
5. Rapid sludge dewatering
6. Easier handling of sludge
7. low levels of microbial growth

Scanned with CamScanner

Fermentation Economies:-

1. Capital investment should be confined to minimum
2. Raw material as cheap as possible
3. Highest Yeilding Strain
4. Saving in labour using automated machine if feaseable
5. Growth cycle as short as possible to obtain highest yeild
6. Simple and Rapid recovery and purification
7. Minimum Effulent Discharge
8. Minimum space requirements to allow high capacity
9. Heat & power used efficiently
10. Safety guidlines and regulations

BEST OF LUCK ☺

REMEMBER ME IN PRAYERS

Scanned with CamScanner